FLEET TEAM DIGEST

737MAX-FTD-34-19002

Issue Title : Automatic Dependent Surveillance-Broadcast (ADS-B) OUT: Navigation Accuracy Category – Velocity (NACv) > 2

Airplane Model	ATA	Minor Model(s)
737MAX	3453-00	-8, -9
Other Model(s)	**ECCN**	
737NG, 777, 767	9E991	

Originated Date	Last Revised Date	Created On
02/15/2019	10/29/2020	10/27/2020
Estimated Completion Date	**Next Update Date**	

Revision Description

Updated Status and Milestones to reflect SB 737-34A3572 release.

Applicability

Airplanes with Collins Aerospace Model GLU-2100 Multi-Mode Receiver (MMR) P/N 822-2532-100 AND either:

ACSS ATC Transponder P/N 9008000-10000, or

Honeywell ATC Transponder P/N 066-01212-0301 installed.

Description

Boeing has incorporated Collins Aerospace Model GLU-2100 Multi-Mode Receiver (MMR) P/N 822-2532-100 into 737 production deliveries beginning in December 2018, and plan on installing this MMR part number on 767 and 777 models in the near-future. The installation of the GLU-2100 MMR may result in two DO-260B ADS-B OUT capable Air Traffic Control (ATC) Transponders transmitting Navigation Accuracy Category - Velocity (NACv) values of 3. The applicable ATC Transponder part numbers, where NACv=3 may be transmitted, when paired with the GLU-2100 MMR, are:

- ACSS P/N 9008000-10000, and

- Honeywell P/N 066-01212-0301

Operators are being informed that aircraft with these MMR/ATC Transponder configurations may not be used in operations requiring NACv>2. At this time, there are NO defined operations anywhere in the world that require a NACv>2, and there are no known near-term plans for such operations. Therefore, there are currently no operational limitations due to the transmission of NACv=3.

NOTE: Installations with DO-260B ADS-B OUT capable ATC Transponder Collins Aerospace P/N 822-1338-205 are NOT affected, since this transponder limits the NACv<3.

Background

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Automatic Dependent Surveillance - Broadcast (ADS-B) is a "surveillance" technology in which an aircraft "depends" on satellite navigation to determine its position, and then "broadcasts" this information "automatically". An ADS-B OUT equipped aircraft broadcasts its position, altitude, velocity, velocity accuracy (NACv), and other aircraft information. This information can be used by Air Traffic Service (ATS) providers for aircraft separation, and by ADS-B IN equipped aircraft, when authorized, to use in a number of different current and future ADS-B IN applications.

Guidance from AC20-165B, “Airworthiness Approval of Automatic Dependent Surveillance – Broadcast OUT", paragraph 3.3.3.7.3 states: “A NACv=3 or NACv=4 should not be set based on GNSS velocity accuracy unless you can demonstrate to the FAA that the velocity accuracy actually meets the requirement”. Furthermore, AC20-165B Appendix B, paragraph B.4.14.3, NACV = 3 or 4, states: “No standard for performance has been developed to support NACv=3 or NACv=4. A NACv=3 or NACv=4 should not be set based on GNSS velocity accuracy unless you can demonstrate to the FAA that the error contributions have been adequately modeled to meet those levels of performance”.

No current airworthiness criteria exists to validate NACv=3 or NACv=4 performance, and there are no near-term applications requiring the use of NACv=3 or NACv=4 performance. As such, airworthiness criteria needs to be developed (e.g. through the FAA Issue Paper process) to validate the performance. Until this criteria is developed and the design verified, ADS-B OUT installations will not be eligible to conduct operations requiring NACv=3 or NACv=4.

Until the airworthiness criteria exists, the FAA has released Memorandum AIR-6B0-18-6B0-DM415, "AC 20-165B General Memo", to allow ADS-B OUT installation approval for installations using GNSS receivers that report NACv=3 performance as long as operators are informed that these ADS-B OUT installations may not be used for operations requiring NACv=3 or NACv=4.

The Collins Aerospace GLU-2100 MMR, in combination with the ACSS or Honeywell ATC Transponder part numbers listed in the Description section above, may transmit NACv=3. Therefore, this FTD is being provided to inform operators with these configurations that operations requiring NACv=3 or NACv=4 are not supported. Operators should also be aware that at this time, there are NO defined operations anywhere in the world that require a NACv>2, and there are no known near-term plans for such operations. Therefore, there are currently no operational limitations due to the transmission of NACv=3.

Status

Boeing is currently working with the FAA to develop an Issue Paper establishing airworthiness criteria for ADS-B OUT NACv=3 performance. Once criteria are established, Boeing intends to coordinate with the FAA to allow for the removal of the operational limitation. The Issue Paper process is expected to be completed by 01-DEC-2020.

A new GLU-2100 Operational Program Software (OPS) part number COL4C-0087-0003, along with Option Selection Software (OSS) part number BCG48-U000-05W9, have been certified. This software limits the Horizontal Figure of Merit (HFOM) output such that NACv>2 will not be output by the GLU-2100. This software update also addresses other GLU-2100 issues documented in 737MAX-FTD-34-19006 and 737MAX-FTD-34-19012.

A Boeing Alert Service Bulletin (SB 737-34A3572) that incorporates OPS part number COL4C-0087-0003 and OSS part number BCG48-U000-05W9 is expected to be available in SEP-2020. Because of the pending availability of Boeing Alert Service Bulletin 737-34A3572, the issue paper is no longer critical.

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SB 737-34A3572 has been released on 15-OCT-2020.

Interim Action

None

Final Action

Operators should install Boeing Alert Service Bulletin 737-34A3572 when available to eliminate the NACV>2 issue and other problems documented in 737MAX-FTD-34-19006 and 737MAX-FTD-34-19012.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Service Bulletin 737-34A3572 Available: 15-OCT-2020 (Complete)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Other	- FAA Advisory Circular AC20-165B		12/07/2015
Other	- FAA Memorandum AIR-6B0-18-6B0-DM415		11/13/2018
Fleet Team Digest Article (FTD)	737MAX-FTD-34-19006		
Fleet Team Digest Article (FTD)	737MAX-FTD-34-19012		
Service Bulletin (SB)	737-34A3572		10/27/2020
Fleet Team Digest Article (FTD)	737MAX-FTD-34-19012		

Related Categories

- Flight Operations : Flight Operations
- Regulatory : Issues mandated by airworthiness directive or other regulatory guidance material

Attachments:

a) FAA Memorandum AIR-6B0-18-6B0-DM415

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